Group Members: Janvi Gandhi, Hannah Yamashita, Hoang Bui (Latent Space Warriors)

Application Area — Food Recommendation / Personalized Nutrition

Task Description — Develop a Bayesian personalized recipe recommendation system that predicts the probability a user will enjoy a recipe based on available ingredients, dietary restrictions, cuisine preferences, cooking time, nutrition, and previous recipe ratings. The system will continuously update its recommendations as users provide new ratings or preferences, allowing recommendations to become more personalized while quantifying uncertainty in each prediction.

Data Type — Structured recipe data (CSV), recipe metadata, ingredients, user ratings, and text descriptions.

https://www.kaggle.com/datasets/shuyangli94/food-com-recipes-and-user-interactions?utm_source=chatgpt.com&select=RAW_recipes.csv

contains:

- *180,000+ recipes*
- *700,000+ user ratings/reviews*
- *Ingredients*
- *Nutrition*
- *Cooking steps*
- *Tags (vegetarian, dessert, Italian, etc.)*
- *User-recipe interactions (perfect for personalization*

How Bayesian recommendation works here:

Suppose a new user signs up.

Initially (the prior), the system doesn't know much:

- *Italian: 30%*
- *Mexican: 25%*
- *Indian: 20%*
- *Chinese: 15%*
- *Other: 10%*

As they interact:

- ⭐⭐⭐⭐⭐ *Pasta → Italian probability increases*
- ⭐ *Curry → Indian probability decreases*
- ⭐⭐⭐⭐⭐ *Pizza → Italian increases again*
- ⭐⭐⭐⭐ *Lasagna → confidence grows*

Instead of saying:

> *"Recommend Pasta"*

system says:

> ***P(user likes Garlic Pasta) = 91%***
> ***P(user likes Veggie Stir Fry) = 64%***
> ***P(user likes Chicken Curry) = 28%***

As users continue rating recipes, the posterior updates and recommendations become more personalized.

Project Title — Bayesian Personalized Meal Recommendation System